

Student Setup & Getting Started Guide

Working with AI APIs • Complete Step-by-Step Instructions • Windows & Mac

Read this BEFORE opening any notebook. Follow every step in order.

What this guide covers:

- Installing VS Code and the Python + Jupyter extensions
- Creating your project folder and opening it in VS Code
- Creating a virtual environment
- Installing the correct Python libraries (OpenAI or Gemini)
- Getting your API key and creating the .env file
- Opening and running the Jupyter notebooks
- Fixing the 10 most common errors



Which API should I use?

Option A — OpenAI (GPT-4o mini): Needs a credit/debit card. Tiny cost (less than \$0.05 today).

Option B — Google Gemini: Completely FREE. Only needs a Gmail/Google account.

If you do not have a card: choose Gemini (Option B).

Setup steps are the same. Only the library installed in Part 5 is different.

PART 1 — Install VS Code

VS Code is a free code editor from Microsoft. It opens Jupyter notebooks, has a built-in terminal, and manages your Python environment — all in one window.

STEP

1

Download and install VS Code

Go to code.visualstudio.com and click the big blue Download for Windows button. Run the downloaded file. Click Next through every screen. Accept all defaults.

💡 **During installation — tick these two extra boxes:**

"Add to PATH (requires shell restart)" — lets you use code command in terminal
"Register Code as an editor for supported file types"

STEP

2

Install the Python and Jupyter extensions

Open VS Code. Click the **Extensions icon** in the left sidebar (looks like 4 squares, or press Ctrl+Shift+X).

Search for and install these two extensions:

- **Python** — by Microsoft (the top result with millions of downloads)
- **Jupyter** — by Microsoft

Click Install on each. Wait for the green tick to appear.

PART 2 — Set Up Your Project Folder

STEP

3

Create your AI Bootcamp folder

Create a new folder called **AI Bootcamp** somewhere easy to find.
Recommended location:

```
D:\AI Bootcamp

# Or on your Desktop:
C:\Users\YourName\Desktop\AI Bootcamp
```

To create it: open File Explorer, navigate to where you want it, right-click any empty space, choose **New → Folder**, name it **AI Bootcamp**.

STEP

4

Put your course files in the folder

Copy these files from your instructor into the **AI Bootcamp** folder:

- **Day3_Gemini_Complete.ipynb** — if using Gemini (free)
- **Day3_OpenAI_Complete.ipynb** — if using OpenAI

Both files are Jupyter notebooks. They open directly in VS Code.

STEP

5

Open the folder in VS Code

Method 1 (easiest):

Right-click your **AI Bootcamp** folder → **Open with Code**

Method 2:

Open VS Code → **File** → **Open Folder** → navigate to AI Bootcamp → click **Select Folder**

☒ **You will know it worked when:**

The VS Code title bar shows "AI Bootcamp"

Your .ipynb files appear in the Explorer panel on the left

PART 3 — Open the Terminal in VS Code

STEP

6

Open the built-in terminal

In VS Code: click **View** in the top menu → **Terminal**

Or press: **Ctrl + `** (the backtick key, top-left corner of keyboard, under Escape)

A terminal panel opens at the bottom. It should already be in your AI Bootcamp folder.

Confirm by checking the path shown:

```
D:\AI Bootcamp>
```

```
# If it shows a different folder, navigate there:  
cd "D:\AI Bootcamp"
```

STEP

7

Switch terminal to Command Prompt (Windows only)

VS Code may open PowerShell by default. Switch to Command Prompt to avoid errors.

Press **Ctrl+Shift+P** → type **Terminal: Select Default Profile** → choose **Command Prompt**

Close the current terminal (click the X on it) and open a new one with the + button.

Why Command Prompt and not PowerShell?

PowerShell blocks script files with: "cannot be loaded because running scripts is disabled"

Command Prompt does not have this restriction.

If you must use PowerShell, run this once to fix it:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

PART 4 — Create Your Virtual Environment

A virtual environment is an isolated Python installation just for this project. It keeps your AI libraries separate from everything else on your computer.

STEP

8

Check Python is installed

In the VS Code terminal, type this and press Enter:

```
python --version

# Expected output:
Python 3.10.0 (or any version 3.8 or newer)

# If that does not work, try:
python3 --version
```

⚠ If you see "python is not recognized":

Python is not installed or not in your PATH.

Download from: python.org/downloads

During install: TICK the box "Add Python to PATH"

After installing: close and reopen VS Code, then try again.

If you have Anaconda installed, use the full path:

C:\Users\YourName\anaconda3\python.exe --version

STEP

9

Create the virtual environment

In the terminal, type exactly (make sure you are in the AI Bootcamp folder):

```
python -m venv venv

# This creates a folder called "venv" inside AI Bootcamp
# Takes 30-60 seconds. Wait for the prompt to return.
```

⚠ If python is not found and you have Anaconda:

Use the full Anaconda path — replace YourName with your Windows username:

C:\Users\YourName\anaconda3\python.exe -m venv venv

STEP

1

0

Activate the virtual environment

Type this in the terminal:

```
# Windows (Command Prompt):
venv\Scripts\activate
```

```
# Your prompt changes — you will see (venv) at the start:  
(venv) D:\AI Bootcamp>
```

```
# Mac or Linux:  
source venv/bin/activate
```

☑ How to know activation worked:

You MUST see (venv) at the start of the terminal prompt.

(venv) D:\AI Bootcamp> ← this is correct

D:\AI Bootcamp> ← this means NOT activated — fix it before continuing.

⚠ Common activation errors and fixes:

Error: "cannot be loaded because running scripts is disabled"

→ You are in PowerShell. Switch to Command Prompt (see Step 7).

Error: "The system cannot find the path specified"

→ The venv folder was not created. Run Step 9 again.

Error: "python is not recognized" when creating venv

→ Use the full Anaconda path shown in Step 9.

STEP

1
1

Tell VS Code to use this virtual environment

Press **Ctrl+Shift+P** → type **Python: Select Interpreter** → press Enter

In the list that appears, look for the option containing **venv** in the path:

```
.\venv\Scripts\python.exe ← click this one
```

If you do not see it, click **Enter interpreter path** → **Find...** → navigate to AI
Bootcamp\venv\Scripts\python.exe

PART 5 — Install Required Libraries

⚠ Check (venv) is showing BEFORE running any pip command:

Your prompt must look like: (venv) D:\AI Bootcamp>

If you do not see (venv): run venv\Scripts\activate first.

Installing without (venv) will install to the wrong Python.

STEP

5
A

Option A: Install OpenAI libraries (if using OpenAI)

Only do this if you are using OpenAI with a credit/debit card.

```
# Remove old conflicting library if present
pip uninstall google-generativeai -y

# Install OpenAI libraries
pip install openai python-dotenv tiktoken

# Also install Jupyter
pip install jupyter ipykernel

# Wait 2-3 minutes. Expected final output:
# Successfully installed openai-... python-dotenv-... tiktoken-...
```

STEP

5
B

Option B: Install Gemini libraries (if using Gemini — FREE)

Do this if you are using Google Gemini (free, no card needed).

```
# CRITICAL: Remove the OLD deprecated library first
# google-generativeai = OLD library = causes AttributeError errors
# google-genai = NEW library = correct, use only this one
pip uninstall google-generativeai -y

# Install the NEW correct Gemini library
pip install google-genai python-dotenv tiktoken

# Also install Jupyter
pip install jupyter ipykernel

# Wait 2-3 minutes. Expected final output:
# Successfully installed google-genai-... python-dotenv-...
# tiktoken-...
```

⚠ The two library names look almost the same but are completely different:

WRONG: pip install google-generativeai (OLD, deprecated, causes AttributeError)

RIGHT: pip install google-genai (NEW, correct, use only this)

STEP

1

Verify the installation worked

2

Run this in the terminal (with (venv) showing):

```
# Test Gemini installation:
python -c "from google import genai; print(genai.__version__)"

# Expected: a version number like 1.75.0 or 2.10.0

# Test OpenAI installation:
python -c "from openai import OpenAI; print('OpenAI OK')"
```

```
# Expected: OpenAI OK
```

If you see ImportError or ModuleNotFoundError:

1. Is (venv) showing? If not, activate it and install again.
2. Did you install google-genai (not google-generativeai)?
3. Close VS Code completely and reopen it, then try again.

Option A — OpenAI Key (requires card)

STEP

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Get your OpenAI API key

1. Open your browser → go to platform.openai.com
2. Click **Sign Up** → use any email → verify your email
3. Click your profile icon (top right) → **API Keys** → **Create new secret key**
4. Name it **AI Bootcamp** → click Create
5. COPY THE KEY IMMEDIATELY — it starts with **sk-proj-** and is very long (~164 characters). You only see it ONCE.
6. Settings → Billing → Add payment method → add your card
7. Settings → Billing → Usage limits → set **Hard Limit = \$5** ← do this BEFORE making any API calls

Option B — Google Gemini Key (completely FREE)

STEP

13

Get your Gemini API key

8. Open your browser → go to aistudio.google.com
9. Click **Sign in** → use any Google account (Gmail works perfectly)
10. In the left sidebar, click **Get API key**
11. Click **Create API key** → **Create API key in new project**
12. COPY THE KEY — it starts with **AIzaSy** and is about 39 characters long
13. No card needed. No charges. This is completely free.

PART 7 — Create Your .env File

The .env file stores your API key safely. Python reads it automatically. Your key never goes directly in your code.

STEP

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4

Create and save the .env file

Step 1: Press Windows key → type **Notepad** → press Enter to open it.

Step 2: Type exactly ONE of these lines (use YOUR actual key, not the placeholder):

```
# If using Gemini — type this line:
GOOGLE_API_KEY=AIzaSy-paste-your-actual-key-here

# If using OpenAI — type this line instead:
OPENAI_API_KEY=sk-proj-paste-your-actual-key-here
```

⚠ Three rules that must not be broken — any mistake means key not found:

Rule 1: No spaces around the = sign. KEY=value NOT KEY = value

Rule 2: No quotes around the value. KEY=abc123 NOT KEY="abc123"

Rule 3: Your real key, not the placeholder text above.

Step 3: Save the file: **File** → **Save As**

14. Navigate to your **AI Bootcamp** folder

15. File name: type **.env** (dot-env, nothing else, no extension)

16. Save as type: click the dropdown → choose **All Files (*.*)** ← THIS IS IMPORTANT

17. Click **Save**

STEP

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5

Verify the .env file is correct

In the VS Code terminal, type:

```
dir

# Look for .env in the list
# You should NOT see .env.txt — only .env
```

Your AI Bootcamp folder should contain:

```
AI Bootcamp/
├── .env                ← your API key (correct)
├── venv/              ← virtual environment
├── Day3_Gemini_Complete.ipynb ← Gemini notebook
└── Day3_OpenAI_Complete.ipynb ← OpenAI notebook
```

⚠ The most common mistake on Windows — .env gets saved as .env.txt:

Windows adds .txt to files even when you type .env

This happens if you forgot to choose "All Files" when saving.

To check: type "dir" in the terminal
If you see .env.txt instead of .env, fix it:
ren .env.txt .env

Then run "dir" again to confirm it now shows as .env

PART 8 — Open and Run the Notebook

STEP

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Open your notebook in VS Code

In the VS Code left panel (Explorer sidebar), click on:

- **Day3_Gemini_Complete.ipynb** — if using Gemini (free)
- **Day3_OpenAI_Complete.ipynb** — if using OpenAI

The notebook opens in a new tab. You will see all the cells inside it.

STEP

1

7

Select the venv kernel

In the top-right corner of the notebook, you will see a button. It might say **Select Kernel** or show a Python version number.

Click it → click **Python Environments** → find the option with **venv** in the path → click it.

```
.\venv\Scripts\python.exe ← select this one
```

⚠ If you do not see venv in the list:

1. Go back to terminal and confirm (venv) is showing.
2. Press Ctrl+Shift+P → Python: Select Interpreter → browse to venv\Scripts\python.exe
3. Try closing and reopening VS Code.

STEP

1

8

Run cells from top to bottom

Click on the first cell in the notebook to select it.

Press **Shift+Enter** to run it. The output appears below the cell.

Run every cell in order from top to bottom.

🔗 Keyboard shortcuts for running cells:

Shift+Enter run this cell and move to the next

Ctrl+Enter run this cell and stay on it

Ctrl+Shift+P → Run All Cells run every cell at once

Cells must run in order. If a cell uses a variable, the previous cell must have run first.

STEP

1

9

What to expect when everything works

After running the setup cell (Topic 3), you should see:

```
# Gemini:
Client ready! We are connected to Gemini.

# OpenAI:
Client created! Ready to talk to the AI.
```

☑ **If you see the above messages: your setup is complete!**

Continue running cells from top to bottom.

Each cell has explanations above it — read them before running.

⚠ **Gemini quota error — 429 with "limit: 0":**

Your account cannot use gemini-2.0-flash. Find the line:

`GEMINI_MODEL = "gemini-2.0-flash"`

Change it to:

`GEMINI_MODEL = "gemini-1.5-flash"`

Then rerun that cell.

To see all models available to your account, run:

```
for m in client.models.list(): print(m.name)
```

PART 9 — Daily Startup Checklist

Every time you come back to this project, do these 4 steps:

1

Open VS Code with your project

File → Open Folder → select AI Bootcamp → Select Folder

2

Open the terminal

View → Terminal (or press Ctrl+`)

3

Activate the virtual environment

Type: `venv\Scripts\activate`

Wait for (venv) to appear at the start of the prompt.

4

Open your notebook and select kernel

Click the .ipynb file in the left panel.

Top-right: click Select Kernel → choose the venv option.

PART 10 — Troubleshooting

Find your error below and follow the fix exactly.

 **AttributeError: module 'google.generativeai' has no attribute 'Client'**

You have the old deprecated library installed.

Run in terminal:

```
pip uninstall google-generativeai -y
pip install google-genai
```

Then in VS Code: click Restart in the top toolbar of the notebook.

 **ImportError: cannot import name 'genai' from 'google'**

Both old and new libraries are installed and conflicting.

Run in terminal:

```
pip uninstall google-generativeai -y
pip uninstall google-genai -y
pip install google-genai
```

Then restart the kernel.

 **ClientError: 429 RESOURCE_EXHAUSTED with limit: 0**

Your account cannot use that model. Find this line in the notebook:

```
GEMINI_MODEL = 'gemini-2.0-flash'
```

Change it to:

```
GEMINI_MODEL = 'gemini-1.5-flash'
```

To see all your available models run:

```
for m in client.models.list(): print(m.name)
```

 **429 RESOURCE_EXHAUSTED (rate limit hit)**

You sent too many requests in one minute. The free tier allows 15 per minute.

Wait 60 seconds then try again.

In any loop that calls the API, add: `time.sleep(3)` between calls.

 **Key not found / os.getenv('GOOGLE_API_KEY') returns None**

Check 1: Is `.env` in the AI Bootcamp folder? Run: `import os; print(os.listdir('.'))`

Check 2: Is it named `.env` and not `.env.txt`? Run `dir` in terminal.

Check 3: No spaces around `=` in the `.env` file?

Check 4: Rename if needed: `ren .env.txt .env`

 **(venv) not showing in terminal**

Run: `venv\Scripts\activate`

If that fails with 'cannot be loaded': you are in PowerShell.

Press Ctrl+Shift+P → Terminal: Select Default Profile → Command Prompt

Open a new terminal and try again.

 **python is not recognized as a command**

Open Windows Settings → Apps → Advanced app settings → App execution aliases.

Turn OFF the toggles for python.exe and python3.exe.

Close and reopen VS Code. Try `python --version` again.

 **ModuleNotFoundError: No module named 'google' or 'openai'**

Library not installed in your venv. Confirm (venv) is showing, then:

```
pip install google-genai python-dotenv tiktoken
```

Then restart the kernel: click the restart icon in the notebook toolbar.

 **Access is denied when running pip**

`pip` is installing to system Python, not your venv.

Make sure (venv) is showing. If still failing, add `--user` flag:

```
pip install google-genai python-dotenv tiktoken --user
```

The `--user` flag bypasses the permission issue.

 **venv\Scripts\activate: UnauthorizedAccess or cannot be loaded**

You are in PowerShell. Run this one-time fix:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Type Y and press Enter. Then try `venv\Scripts\activate` again.

Or switch to Command Prompt (Ctrl+Shift+P → Terminal: Select Default Profile).

You are ready! Open your notebook and start with Cell 1.

Still stuck? Raise your hand in class or message your instructor.